Nicolaos Petropoulos

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https://nicolaosp.github.io/Website/

Education

Jan. 2025 - Apr. 2026 — **Post-doctoral fellow** in the Center for Turbulence Research — Stanford University

- Developed a novel theoretical framework for stirring and mixing in stratified turbulence by using tools from statistical mechanics and stochastic processes.

Oct. 2021 - Sept. 2024 — **Ph.D.** candidate in Applied Mathematics and Theoretical Physics — DAMTP, Cambridge
Marie Skodowska-Curie Early Stage Researcher Fellow
Supervisor: Prof. Colm-cille P. Caulfield

- Using theory, laboratory experiments, simulations, machine learning and observations, I developed a stochastic approach to transport, dispersion and mixing in the ocean, i.e. the way turbulent flows disperse and mix tracers such as heat, salinity, chemicals or plastics.
- Engineered a data pipeline processing petabytes of simulation and observational data to uncover the geometry of ocean mixing.
- Work presented at international conferences and awarded a **Smith-Rayleigh and Knight-Rayleigh Prize** and a **UK Fluid Network Best Thesis Prize.**

June 2024 - Aug. 2024 — **Fellow** of the Geophysical Fluid Dynamics Summer Program — WHOI, Woods Hole

- Selected for one of 10 fully funded fellowships from a worldwide pool of applicants.
- Developed a new model for the settling dynamics of complex porous and elastic objects, relevant to biological and industrial particle transport.

2020 - 2021 — **Part III** of the Mathematical Tripos (Master of Advanced Study) — DAMTP, Cambridge

- Theoretical courses in Pure Mathematics (Analysis of Partial Differential Equations and Stochastic Differential Equations, Stability Analysis, Non-linear Analysis) and Applied Mathematics (Stochastic Processes in Biology, Biological Physics, Fluid Dynamics of Climate, Hydrodynamic Stability).

2017 - 2020 — **Master** of engineering curriculum — Ecole Polytechnique, Paris

- Theoretical courses: Applied Mathematics, Fluid Mechanics, Probability Theory, Statistical Physics, Quantum Mechanics, Relativity.
- Focus areas: Dynamical Systems, Control Theory, Numerical Methods, Variational Methods, Computational Fluid Dynamics.
- Developed an autonomous boat: project rewarded by school jury.
- Developed software for the autonomous detection of fishing offences using AIS data.
- Developed a finite element solver from scratch.

2014 - 2017 — **Bachelor** equivalent program
(Classes Préparatoires aux Grandes Ecoles) — Lycée Louis le Grand, Paris

- Ranked 7th at Ecole Polytechnique's entry exams, among more than 2,000 candidates.

Work Experience

Sept. 2026 - Oct. 2026 — **Research scientist on the Falkor (too)** — Schmidt Ocean Institute

- Part of the research staff of the 'Life at the Mid-Atlantic Ridge fracture zone' research cruise.
- Developed a GPU-accelerated regional numerical model that simulates the fluid dynamics and biogeochemistry of the Equatorial Atlantic ocean.

March 2020 - Aug. 2020 **Research internship** LATMOS - Sorbonne University
Supervisors : Dr. Ludivine Oruba (LATMOS) and Prof. Emmanuel Dormy (ENS Paris)

- Built a Lagrangian numerical solver for the wave action equation to efficiently simulate energy transfers between tropical cyclones and the ocean.

June 2019 - Aug. 2019 **Research internship** Emirates Team New Zealand (Auckland, New Zealand)
Supervisors : Thomas Khyn (Software and Simulation Engineer) and Dr. Dan Bernasconi (Head of Design)

- Three months with the Design Team of the Defender of the 36th America's Cup.
- Developed a ride-height autopilot using evolutionary algorithms, ensuring stable autonomous foiling of the sailing boat.

Skills

Languages English: Fluent, French & Greek: Native language.
Programming Python, PyTorch, Julia, Wolfram Mathematica, HPC programming on the Oak Ridge National Laboratory supercomputers.

Publications

1. **Nicolaos Petropoulos**. 'The routes towards homogeneity in stratified turbulent flows' *Journal of Fluid Mechanics* 1032 (2026): R2.
2. **Nicolaos Petropoulos**, Stephen M. de Bruyn Kops, Patrice Meunier, Emmanuel Villermaux. 'Lyapunov exponents in conservative systems: the importance of rare compression events' *Submitted, PNAS*.
3. **Nicolaos Petropoulos**. 'Nondissipative variable density turbulence: routes toward statistical equilibrium' *Center for Turbulence Research, Annual Research Briefs*, 2025.
4. **Nicolaos Petropoulos**, Lois E. Baker, Miles M. P. Couchman, Ali Mashayek, Colm-cille P. Caulfield. 'Potential biases in marginal instability and overturn-based approaches to the parameterisation of ocean mixing' *In prep., Geophysical Research Letters*.
5. **Nicolaos Petropoulos**, Stephen M. de Bruyn Kops, Colm-cille P. Caulfield. 'Modelling dispersion in stratified turbulent flows as a resetting process' *Journal of Fluid Mechanics* 1002 (2025): A48.
6. **Nicolaos Petropoulos**, Miles M. P. Couchman, Ali Mashayek, Stephen M. de Bruyn Kops and Colm-cille P. Caulfield. 'Prandtl number effects on extreme mixing events in forced stratified turbulence' *Journal of Fluid Mechanics* 983 (2024): R1.
7. **Nicolaos Petropoulos**, Colm-cille P. Caulfield, Patrice Meunier, Emmanuel Villermaux. 'Settling versus mixing in stratified shear flows' *Journal of Fluid Mechanics* 974 (2023): R1.
8. **Nicolaos Petropoulos**, Ali Mashayek, and Colm-cille P. Caulfield. 'Turbulent disruption of density staircases in stratified shear flows.' *Journal of Fluid Mechanics* 961 (2023): A30.